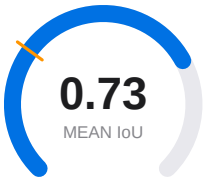
What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

FINDING

419210002 is the best physical fit.

Forward replay tested 10 candidates; the runner-up fits at 0.49. A ranking by physical consistency under tested assumptions — an investigative lead for a human analyst.



MEAN IOU	VARIANTS WON	LEAD RETAINED	CANDIDATES
0.73	**6** / 6	**84%**	**10**
the fit gate is 0.3	forcing variants where it fits best	of 80 tested changes	replayed by forward replay

EVIDENCE

- Best fit under **6 of 6** forcing variants
- Inside the origin area **7 h** before the image
- Ahead of the runner-up by a mean loss gap of **0.0975**
- The nearest vessel, **419245446**, 0.0045° away, is not supported

Candidates, ranked by physical fit

Each candidate's release replayed forward under six forcing variants and compared with the observed slick.

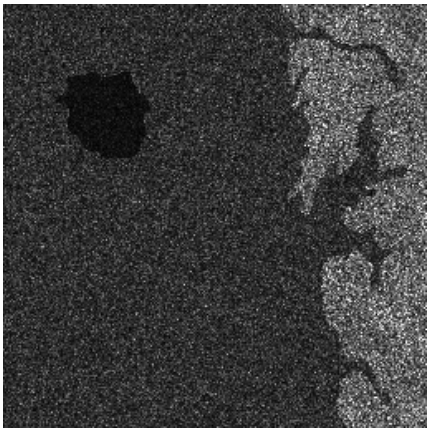
#	VESSEL	MEAN IoU		VARIANTS WON	RELEASE STARTS
1	419210002		0.727	6 of 6	9 Sep 2026, 22:30 UTC
2	419105044		0.487	0 of 6	9 Sep 2026, 21:00 UTC
3	419737493		0.365	0 of 6	9 Sep 2026, 12:30 UTC
4	419364347		0.320	0 of 6	9 Sep 2026, 14:30 UTC
5	419433516		0.307	0 of 6	9 Sep 2026, 04:30 UTC
6	419245446		0.189	0 of 6	10 Sep 2026, 02:30 UTC
7	419925490		0.004	0 of 6	10 Sep 2026, 02:30 UTC
8	419761151		0.000	0 of 6	9 Sep 2026, 21:30 UTC

#	VESSEL	MEAN IoU	VARIANTS WON	RELEASE STARTS
9	419634099	0.000	0 of 6	9 Sep 2026, 21:30 UTC
10	419551783	0.000	0 of 6	9 Sep 2026, 17:30 UTC

The proximity baseline — the vessel nearest the slick when the image was taken — would pick **419245446**, ranked 6 by forward replay.

The scene

The radar image the slick was detected in. Radar darkness alone is not proof of oil.



Acquired	10 Sep 2026, 04:30 UTC
Sensor	synthetic SAR-like (VV proxy, single channel)
Pixel spacing	433.59 m
Data status	synthetic
Label source	generator ground truth (synthetic) — not a human or independently confirmed label
Projection	EPSG:32643

Robustness

Does the leader survive small changes to the model's assumptions?

84%
LEAD RETAINED

80 tested changes to the model's assumptions; **13** changed the leader.

The smallest change that flips it: **along-shore current bias (m/s) -0.03, diffusivity multiplier 0.6** (magnitude 1.0), under which **419433516** leads.

Single-axis and two-axis perturbations only. Three-or-more-axis combinations were not tested, so absence of a flip is evidence of stability across the tested set, not a proof of stability.

Forward exposure

Where the oil is modelled to drift after the image. Reanalysis-style retrospective field — not an operational forecast.

AFTER THE IMAGE	MODELLED SPREAD	SHORELINE	FISHING GROUND	MANGROVE RESERVE
+6 h 10 Sep 2026, 10:30 UTC	491 km^2	0%	42%	0%
+12 h 10 Sep 2026, 16:30 UTC	541 km^2	0%	40%	0%
+24 h 11 Sep 2026, 04:30 UTC	585 km^2	1%	28%	0%

Traffic filtering

Every vessel in the scene, checked against where the oil could have come from over the 24 hours before the image.

10	Candidate	track data present and inside the backward envelope at some queried time
1	Excluded	track data at every queried time, never inside the envelope

Unassessed is reported separately from excluded: missing evidence is not exoneration.

Coverage-aware matching

Matching radar contacts to AIS when some AIS is hidden — tested on synthetic contacts.

RATE	VESSELS HIDDEN	AREA GOES DARK
Hidden AIS recovered	**86%** 54/63	**84%** 31/37
False alerts	**34%** 48/142	**35%** 28/80
Mistaken promotions	**22%** 9/40	**33%** 7/21

Provenance

AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.
Generated	26 Sep 2026, 17:11 UTC
Software	Python 3.12.14 · Darwin 25.2.0 (arm64)

File integrity

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

attribution_result.json	`28f6266e0be581573ba7eefb4866984cad70433494d3e1d43a09f6767cb90cbb`
coverage_result.json	`1e82a0203e8eba1ef3b73f7a8ac1aa0161af75cbe18112684421f45f1a07c78f`
forcing.npz	`4b4a3d7409108fc9ff8a0bc13aebea75a693c55c3f7b0ee4d714330907fec93b`
forecast_result.json	`360170a11e8fab91b86c41e483819a5829e467dfedb2aa4eaf70f8a7bc88aa70`
fragility_result.json	`c982850c8bf6dcf26f9aef30516cce58b2983bed184a051345312ac2cb2fd33b`
historical_mapping.json	`0e59d782ef6b8cbbab0b7da9547edca91c732e25b92af128a0fe07691278ccef`
manifest.json	`26de0a4d06b4bc8030568407c00719358354bff8215d985ed4ac416d7d4e556c`
mask_groundtruth.png	`744117fc89bfd9722f97be26b98be067ec86edcb2b4fd13467920b39749fe833`
naive_baseline_result.json	`98441527f8e3dc37b61584d507b3f110eee841cc0ba269e6e7c3d76d0eec43b5`
next_observation_result.json	`29f9f10a8c7b7dd6fc1ebdecee306461a5fdbbdf109704c9a0678fed1277e540`
scene.png	`b01fa9684715b91309fcd557edc5a8389d9b9a615fd801e3080f4f543ed6a958`

Uncertainty

— Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

— Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

— Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

— Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

Restated. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.